

```
[14]: import pandas as pd
import numpy as np
```

```
[2]: # Imaginary Sales Data for 1st and 2nd Quarters for Global Company
q1 = {'Japan': 80, 'China': 450, 'India': 200, 'USA': 250}
q2 = {'Brazil': 100, 'China': 500, 'India': 210, 'USA': 260}
```

```
[5]: sales_q1 = pd.Series(q1)
sales_q2 = pd.Series(q2)
```

```
[6]: sales_q1
```

```
[6]: Japan      80
      China    450
      India    200
      USA      250
      dtype: int64
```

```
[7]: sales_q2
```

```
[7]: Brazil     100
      China     500
      India     210
      USA       260
      dtype: int64
```

```
[9]: print(sales_q1['Japan'])
```

```
80

Now numerically, use iloc
```

```
[11]: print(sales_q1.iloc[0])
```

```
80

To check what keys are available for us inside a series:
```

```
[12]: sales_q1.keys()
```

```
[12]: Index(['Japan', 'China', 'India', 'USA'], dtype='str')
```

```
[12]: Index(['Japan', 'China', 'India', 'USA'], dtype='str')
```

We can perform broadcasted operations on them

```
[13]: [1,2] * 2
```

```
[13]: [1, 2, 1, 2]
```

We see that, in list, it got increased, but not doubled (the values), so in NumPy arrays:

```
[15]: np.array([1,2]) * 2
```

```
[15]: array([2, 4])
```

Now we see that the operation got broadcasted so we got double of the array

So, in Pandas series, it's 'broadcasted operations' so:

```
[16]: sales_q1 * 2
```

```
[16]: Japan      160
      China     900
      India     400
      USA       500
      dtype: int64
```

We see that the values got doubled, unlike lists - also any other operations

We will do sales_q1 + sales_q2 but notice that Japan is replaced by Brazil in sales_q2 so:

```
[17]: sales_q1 + sales_q2
```

```
[17]: Brazil      NaN
      China    950.0
      India    410.0
      Japan      NaN
      USA      510.0
      dtype: float64
```

Now, to avoid NaN and fill up with some value, do this:

```
[20]: sales_q1.add(sales_q2, fill_value= 0)
```

```
[20]: Brazil     100.0
      China    950.0
      India    410.0
      Japan      80.0
      USA      510.0
      dtype: float64
```

Now, we see that Brazil and Japan got its values

```
[ ]:
```